

ASSIGNMENT – 3

Echo Server Using TCP

AIM:

To develop a socket program to simulate an Echo Server, demonstrating reliable two-way communication between a client and a server over a TCP connection.

SOURCE CODE:

```
import socket

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("127.0.0.1", 5000))

server.listen(5)
print("Server is waiting for connection...")

conn, addr = server.accept()
print("Connected by : ", addr)

while True:

    data = conn.recv(1024).decode()

    print("Client Says : ", data)
    if data.strip().lower() == "bye":
        print("Client ended chat")
        break

    string = input("Enter your message to client : ")

    conn.send(string.encode())

conn.close()
server.close()
```

```
import socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

client.connect(("127.0.0.1", 5000))

print("Connected to server")
```

```
while True:

string = input("Enter your message to server : ")
client.send(string.encode())

data = client.recv(1024).decode()
print("Server says : ", data)
if data.strip().lower() == "bye":
print("Client ended chat")
break

client.close()
```

OUTPUT:

```
PS C:\Users\srikrishna\Networks Laboratory\exp1> python server.py
Server is waiting for connection...
Connected by : ('127.0.0.1', 55202)
Client Says : Hi this is client
Enter your message to client : hi this is Server
Client Says : Thanks for accepting my connection request
Enter your message to client : It's my pleasure
Enter your message to client :

PS C:\Users\srikrishna\Networks Laboratory\exp1> python client.py
Connected to server
Enter your message to server : Hi this is client
Server says : hi this is Server
Enter your message to server : Thanks for accepting my connection request
Server says : It's my pleasure
Enter your message to server :
```

Learning Outcomes

After completing this experiment, I was able to:

- Understand the working of TCP socket communication between a client and a server.
- Develop a TCP-based client-server application using socket programming.
- Establish and manage reliable connections between client and server.
- Transmit and receive data over a TCP connection.
- Implement an Echo Server, where the server receives data from the client and sends the same data back.